

Karl Hill

SOFTWARE ENGINEERING LEADER | CLOUD-NATIVE PLATFORMS • DEVSECOPS •
MISSION-READY DELIVERY

Summary

Software engineering leader with 20+ years of experience building secure, cloud-native platforms for aerospace, NASA, federal, and mission-critical environments. Combines hands-on architecture, DevSecOps governance, platform engineering, Agile delivery leadership, and stakeholder alignment to improve how teams build, test, release, and operate mission-ready software.

Selected Leadership Impact

- Lead cloud-native mission software delivery across aerospace, NASA, federal, and high-assurance environments.
- Improve engineering discipline through CI/CD, release governance, PR standards, and Definition of Done practices.
- Modernize platforms supporting disaster response, Earth science, telemetry, algorithms, and high-volume public systems.
- Bridge technical execution and stakeholder alignment across engineering, product, operations, and mission teams.

Professional Experience

Staff Aerospace Software Engineer, Jacobs, Chantilly, VA

SEPT 2025 – PRESENT

- Lead cross-functional engineering execution across cloud-native mission services, CI/CD pipelines, multi-repo delivery, release readiness, and integration risk management.
- Guide team execution, onboarding, coaching, and delivery discipline while strengthening engineering quality, ownership, and accountability.
- Coordinate multi-environment release readiness across isolated and integrated systems, reducing late-stage integration risk and improving baseline alignment and traceability.
- Establish and enforce engineering standards — repo and branch governance, PR/review practices, Definition of Done, and documentation — to reduce integration risk across multi-repo and multi-environment systems.
- Drive DevSecOps and release governance through CI/CD, automated testing, quality gates, and security checks to shorten feedback loops and improve system reliability.
- Translate mission and operational needs into roadmaps and sequenced delivery plans, communicating status and tradeoffs clearly to technical and non-technical audiences.

Lead Software Engineer, SSAI / NASA Goddard Space Flight Center, Greenbelt, MD

DEC 2017 – SEPT 2025

- Architected NASA's cloud-based Flood Mapping System on AWS, delivering near real-time, satellite-derived flood products to support disaster response.
- Built a high-performance file and metadata platform on Ceph, improving virtual directory mapping and accelerating discovery and access for large scientific datasets.

Details

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Links

<https://www.linkedin.com/in/khill/>
<https://github.com/karlhillx>
<https://karlhill.com>

Areas of Expertise

Engineering Leadership & Team Development

Cloud-Native Mission Systems

Platform Engineering & Developer Experience

DevSecOps, CI/CD & Release Governance

Secure SDLC & Engineering Standards

Architecture, APIs & Systems Integration

Multi-Environment Delivery & Integration Risk

Agile Delivery, Planning & Stakeholder Alignment

Observability, Reliability & Operational Readiness

- Rebuilt NASA Earth Observatory's high-traffic web platform, supporting ~1.5M monthly visitors while improving performance, UX, and SEO.
- Delivered an automated content registry workflow, boosting data collection efficiency by ~60% and accelerating researcher access to new datasets.
- Implemented DevSecOps delivery with GitLab CI/CD, Docker, and Kubernetes, enabling automated deployments, repeatable releases, and reliable stakeholder approvals.
- Led Agile delivery across teams and stakeholders, translating mission needs into roadmaps, sequencing dependencies, and driving consistent delivery.

Sr. Software Engineer, InformedDNA, St. Petersburg, FL

JAN 2016 – DEC 2017

- Architected and delivered a Laravel-based case management platform, reducing operational costs by **\$30K/year**.
- Led CRM enhancements that improved retention and contributed **~15% revenue growth** through better lifecycle workflows and reporting.
- Built and optimized APIs to improve database efficiency, reduce operational overhead, and increase application performance and reliability.
- Spearheaded platform upgrades and security process improvements, **doubling incident response efficiency** and strengthening operational readiness.

Sr. Software Engineer, Ticomix, Inc., Washington, D.C.

JUN 2012 – MAR 2015

- Delivered SugarCRM solutions for **20+ clients** (including Virginia Department of Transportation, Washington Redskins, Kastle Systems), improving sales operations, visibility, and team productivity.
- Drove execution discipline to **cut backlog ~90%**, improving delivery predictability, product quality, and customer satisfaction.
- Built a lightweight opportunity-identification application that helped customers surface and prioritize new business opportunities.
- Collaborated effectively on distributed teams—clear async communication, strong documentation, and reliable cross-time-zone delivery.

Earlier Engineering Experience

Sabre Corporation • Dante Inc. • Visitar Inc. • Verizon Business

1997 – 2012

- Held software engineering and principal engineering roles across travel, enterprise CRM, telecommunications, managed security services, and large-scale client platforms.
- Built Java, PHP, CRM, telephony, and internal operations systems for enterprise clients including Comcast, Mastercard, Verizon/MCI, and global travel customers.
- Delivered production systems, workflow automation, client onboarding platforms, API integrations, and reliability improvements.
- Helped mature engineering practices through Scrum adoption, testing discipline, code quality, junior developer mentoring, and cross-functional delivery.

Education

Bachelor of Science, Computer Science coursework, University of Maryland

Associate of Arts, General Studies, Howard Community College

Project Management, Rutgers University

Certifications

- Professional Scrum Master™ II (PSM II), Scrum.org
- Professional Scrum Product Owner™ I (PSPO I), Scrum.org
- Professional Scrum Developer™ I (PSD I), Scrum.org
- Certified ScrumMaster® (CSM®), Scrum Alliance

Technical Leadership, Platforms & Engineering Stack

Languages: Python, TypeScript, Java, Bash

Cloud & Platform: AWS, Docker, Kubernetes, Helm/OCI, cloud-native architecture

DevSecOps & Release Engineering: Bitbucket, GitLab CI, GitHub Actions, CI/CD, release automation, quality gates

API & Systems Integration: REST/JSON, OpenAPI/Swagger, service integration, distributed system patterns

Web & Application Platforms: Node.js, React, Vite, Tailwind, Laravel/PHP

Data & Search Platforms: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch/OpenSearch

Engineering Operations: Git, Jira, Confluence, documentation, PR/review workflows, engineering standards

AI-Enabled Engineering: Cursor, GitHub Copilot, Claude, LLM-assisted development, code review, documentation, and engineering automation